Explainable-AI Evaluation for Construction-Safety Vision-Language Models

A Two-Week Feasibility Pilot and the Case for Scale-Up

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Abstract

This report documents a completed feasibility pilot testing whether Professor Abdallah's six-metric explainable-AI (XAI) evaluation framework — Descriptive Accuracy, Sparsity, Stability, Efficiency, Robustness, and Completeness, originally built and validated on tabular network-intrusion-detection models — transfers to a Vision-Language Model (VLM) making visual safety judgments on construction sites. Using Florence-2-base-ft (231M parameters, a single RTX 3070) on 163 images from the ConstructionSite dataset, all six metrics ran end-to-end and produced genuine, interpretable signal. **The framework transfers, but required two real adaptations, not zero changes.** The pilot also surfaced a small number of cross-cutting mechanisms — most importantly, that on Florence-2 the safety judgment is ultimately computed by hand-written geometric code rather than by the model, which means some metrics partly measure the proxy rather than the model. This report presents the method, results, findings, and honest limitations, and motivates a phased scale-up whose central goal is to make the metrics measure *model faithfulness*.

1 Executive Summary

The question. Does a six-metric XAI evaluation framework built for tabular models transfer to a VLM making visual judgments? **The answer: yes, with two quantified adaptations.**

1. **Florence-2 has no native question-answering token.** Each safety rule was reframed as an open-vocabulary grounding query (detect "worker" and the relevant safety object, then apply a per-worker coverage check) rather than a literal yes/no question.
2. **The first version of that proxy barely worked** — 3.0% sensitivity, effectively a constant "compliant" classifier. A person-relative redesign, mirroring the dataset's own per-worker labels, produced a **9× sensitivity gain to 27.0%**.

With both adaptations in place, the six metrics produced the headline numbers in Table 1. The findings that matter more than any single number are cross-cutting: one region-ranking choice weakens three metrics at once on the PPE rules; a stable answer can hide a fully-drifted explanation; and, most consequentially for the next phase, the model does not actually make the safety judgment — our geometric code does.

2 Motivation and Research Question

Professor Abdallah's XAI framework evaluates black-box explanations along six axes and was validated on tabular network-intrusion-detection DNNs, where a "feature" is a scalar column. Construction-safety monitoring with VLMs is a very different setting: the input is an image plus a text prompt, and an "explanation" is a spatial region rather than a feature weight. The research

Metric	Result (163 samples)
Descriptive accuracy (masking the top region flips the answer)	36.2%
Visual sparsity (top-5-of-576-cell attention mass)	0.108
Stability (3-run sampled answer agreement)	77.5%
Robustness, Level 1 (survives blur / low-light / occlusion / contrast)	80.7%
Bounded completeness (top region masking is load-bearing)	43.6%
Efficiency (full six-metric pipeline @ n=1,000)	∼0.5 GPU-hours

Table 1: Headline results. All six metrics ran end-to-end and produced interpretable signal.

question is whether the framework's *structure* survives that move — and, where it does not transfer for free, what minimal adaptation makes each metric meaningful again. A feasibility pilot is the right instrument for this: it surfaces the adaptations a full study would otherwise discover the hard way.

3 Dataset and Task

The pilot uses the ConstructionSite dataset (10,013 images; a 3,004-image test split was used). Each image carries annotator-written violation fields for four safety rules:

Rule	Hazard class	What it checks
rule_1	ppe_violation	Basic PPE present (hard hat / hi-vis vest)
rule_2	fall_hazard	Fall protection (harness / lanyard) at height
rule_3	fall_hazard	Edge protection (guardrail) at height / excavation edges
rule_4	struck_by_risk	Worker inside an excavator's blind spot / swing radius
(none)	compliant	Every applicable safety requirement satisfied

Table 2: The four safety rules and the hazard classes they map to.

A balanced 163-image subset was drawn (50 each for three classes; `struck_by_risk` capped at 13 — a labeling artifact discussed in Section 9). Ground truth comes from the dataset's own annotations; the model is then tested on whether it reaches the same judgment, and whether the region it used is a trustworthy explanation.

4 Model and the Grounding-Proxy Adaptation

Florence-2-base-ft was chosen as a lightweight, region-native baseline: it runs on an 8 GB consumer GPU and produces bounding boxes directly, so a candidate "explanation" (a box) falls out of the model without a separate localization step. Its limitation is central: **it has no free-form question-answering task token**. A safety question therefore cannot be asked literally.

The adaptation converts each rule into open-vocabulary detection calls and derives the answer geometrically. For a PPE rule: detect "worker" boxes and "hard hat" boxes, then mark the scene *compliant* only if every detected worker has a nearby safety-object box (Figure 1). A first version that merely asked "does a hard hat exist anywhere in the image?" failed at 3.0% sensitivity, because multi-worker scenes almost always contain *some* hard hat. The person-relative redesign fixed this (27.0% sensitivity).

Figure 1: The grounding-proxy adaptation working: a "hard hat" query correctly localized on a worker's head. This mechanism underlies the 3.0%→27.0% sensitivity gain.

A caveat that shapes the entire scale-up: Florence-2 never makes the safety judgment. It only locates objects; a hand-written geometric rule (per-worker coverage within a fixed pixel-distance threshold) produces the compliant/violation answer. Several metrics therefore partly measure the sensitivity of that geometric proxy rather than the model's own reasoning — the single most important reason to add a native question-answering model in the full study (Section 10).

5 System Architecture

The pipeline is a sequence of stages that each build on the last: environment setup → balanced sample selection → baseline grounding inference → region standardization (rank candidate boxes into an ordered list and provide a masking primitive) → the five per-sample metrics plus the efficiency accounting → bounded completeness (a roll-up over two earlier stages) → a cross-metric summary. Every metric consumes the region-standardization layer, which is why one choice made there (how to rank candidate regions) propagates into several metrics at once — a fact that becomes the pilot's central cross-cutting finding.

6 The Six Metrics: Method and Results

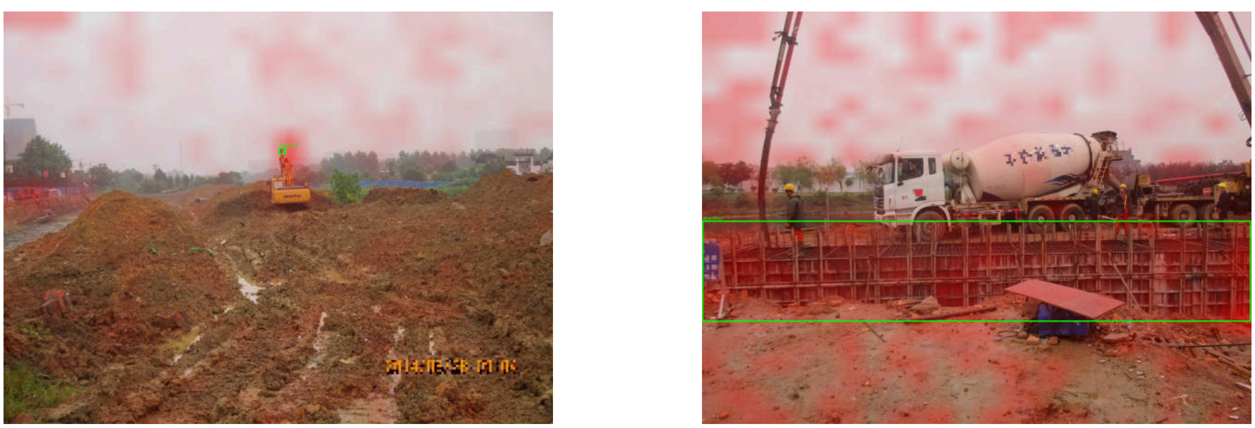
6.1 Descriptive Accuracy — does masking the top region change the answer?

Mask the top-ranked explanation region, re-run, and check whether the answer flips; a genuinely load-bearing region should flip it. Headline: **36.2%** (top-1), 35.0% (top-1+2). Figure 2 shows a clean case: masking the excavator box in a proximity rule removes half the proximity comparison and flips `hazard`→`safe`. The metric's main weakness is that it is confounded by the proxy it sits on: rules whose proxy barely discriminates have little for masking to disrupt.

Figure 2: Descriptive accuracy behaving as designed: masking the top-ranked region (here the excavator in a struck-by-risk sample) changes the model's judgment for a reason that tracks the safety concept.

6.2 Visual Sparsity — how concentrated is the attention?

Because the planned attention-rollout method assumes a single-modality encoder and Florence-2 fuses image and text tokens in one shared self-attention stack, rollout was verified inapplicable and replaced with decoder→encoder cross-attention. Headline top-5-of-576-cell mass ratio: **0.108**. The central finding is a *confound*: concentration correlates −0.70 with object size — a small hard hat looks "focused" simply because it covers few cells (Figure 3). Cross-rule sparsity comparisons are invalid without a size control.



(a) Small hard hat: a sharp, concentrated hotspot.

(b) Large guardrail: attention spread across the frame.

Figure 3: The size confound made visible: concentration tracks object footprint, not explanation quality.

6.3 Stability — does the explanation stay put across reruns?

Deterministic reruns would be trivially 100% stable, so stability is measured under stochastic decoding, confirmed non-vacuous (answer agreement 0.775, not 1.0). Headline object-box overlap **0.602**. The finding: a healthy-looking top-region overlap (0.667) *hid* the instability of the actual safety object (rule_1 hard hat only 0.499 vs. rule_4 excavator 0.922). Small, body-worn PPE items are least stable — partly a genuine effect and partly the same size bias, since IoU penalizes small boxes. Figure 4 overlays three reruns' top boxes.

Figure 4: Three stochastic reruns overlaid. The large top-ranked (worker) boxes sit almost exactly on top of each other; the small hard-hat box the rule is actually about is far less stable underneath.

6.4 Robustness — do answer and explanation survive perturbation?

Four image perturbations plus prompt rewording, under deterministic decoding to isolate perturbation from sampling noise. Level-1 answer survival: **80.7%**. **This chapter contains the pilot's strongest single result:** in 28.3% of cases where the final answer did *not* change, the model's evidence box nonetheless drifted to something unrelated (Figure 5). An answer-level metric alone would certify these as perfectly robust. This is direct evidence that answer-level metrics are necessary but not sufficient — and it motivates reporting answer-level and explanation-level robustness as two never-blended numbers.

6.5 Bounded Completeness — is the top region actually necessary?

A necessity test: does removing the top region force the model to change its answer? Headline **43.6%** supported. The metric is built from the descriptive-accuracy signal (so the two are not independent confirmations), and its by-rule pattern is the clearest measurement of the area-ranking confound (Section 7). The pilot also ran an extra 159-rerun check that revealed a small inflation from a fallback artifact (Figure 6), correcting the number from 43.6% to 41.7% — an example of the rigor a full study should standardize.

6.6 Efficiency — is the pipeline affordable at scale?

The full six-metric pipeline costs ~**0.5 GPU-hours at** n=1,000 on a single RTX 3070 (Figure 7). Compute is conclusively *not* the scaling obstacle; every other issue in this report is about validity,

(a) Before: the worker's evidence box, fully visible.

(b) After occlusion: answer unchanged, but the evidence box relocated almost entirely (IoU 0.005).

Figure 5: The centerpiece finding: a stable answer hiding a fully-drifted explanation.

(a) Before: the "worker" box and "harness" box are nearly pixel-identical.

(b) After masking the top region: both are erased at once; the answer flips via a fallback, not via genuine necessity.

Figure 6: The worker-fallback artifact that inflates completeness, traced to a specific case.

not cost. (The one caveat: this budget is for the current pipeline; adding gradient-based attribution or dense detection in the full study would change it.)

7 Cross-Cutting Findings

Reading the metrics together reveals that several "separate" weaknesses are one story:

1. **One region-ranking choice weakens three metrics at once.** Candidate regions are ranked by box area as a stand-in for importance (no true saliency signal exists). Because a worker's body box is almost always larger than the PPE item, the worker is ranked first in 96% of PPE samples — so descriptive accuracy, stability, and completeness all end up testing the worker, not the hard hat, precisely on the rules the project cares about most. This is the single highest-leverage fix.

2. **A fallback branch recurs across three tests.** Masking or perturbing the worker can make it undetectable, rerouting the proxy into a degenerate scene-level branch that flips the answer for reasons unrelated to the safety concept. The same mechanism appears under deliberate

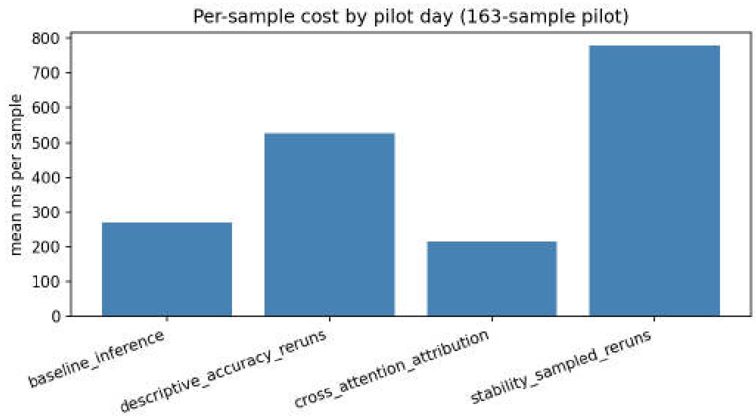


Figure 7: Per-sample inference cost by pipeline stage. Even summed and extrapolated, the full pipeline is well under an hour of GPU time per 1,000 samples.

masking, under perturbation, and inside the completeness check — found and quantified each time, not left as noise (Figures 6, 8).

3. **Object size confounds two metrics.** Both sparsity and stability partly measure object footprint rather than explanation quality, because attention-concentration and IoU are both size-sensitive. This must be controlled before comparing PPE rules to large-object rules.

4. **The model does not make the judgment.** Florence-2 only returns boxes; a geometric rule produces the answer. So some metrics measure the proxy, not the model — the deepest reason to add a native-VQA model in the full study.

(a) Before: both baseline detections are already wrong but geometrically satisfy the check.

(b) After occlusion: both detections vanish; the answer flips via the fallback, not via genuine sensitivity to a covered region.

Figure 8: The fallback-rerouting mechanism triggered by perturbation rather than by deliberate masking.

7.1 Where all six metrics land, side by side

Figure 9 is the cross-metric summary: five per-sample metrics broken down by hazard class on their own native scales. It is the single figure that most compactly carries the pilot's quantitative content, and the source the cross-cutting findings were read from.

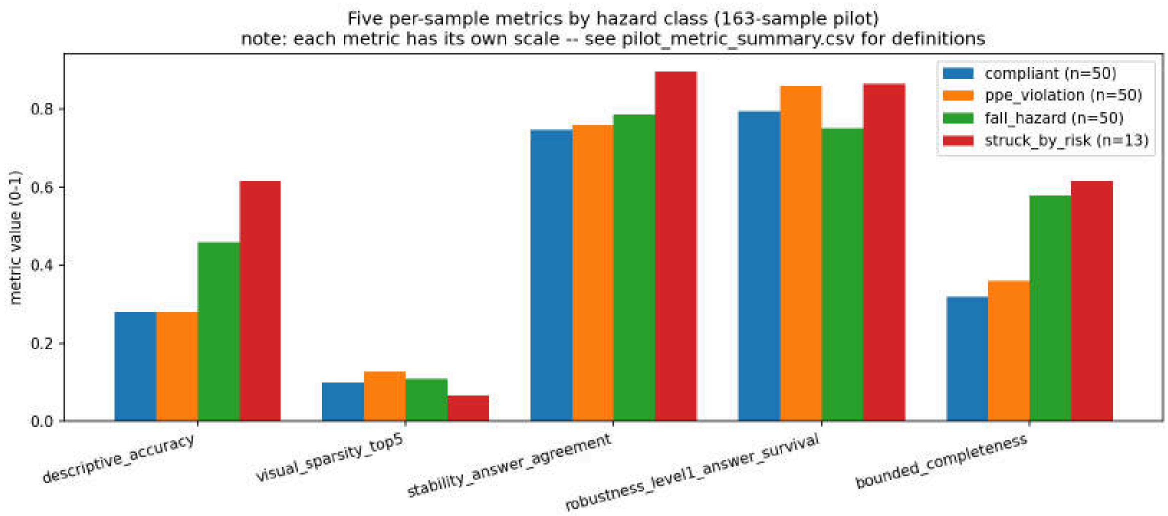


Figure 9: Cross-metric summary by hazard class. `struck_by_risk` rests on n=13 and should be read as directional only (see Section 9).

8 Reproducibility

The full pipeline was re-run from a clean checkout in a brand-new virtual environment on a fresh 20-sample subset, with zero manual intervention — every step exited cleanly. The check did more than confirm reproducibility: because it ran at a different sample size than development, it exposed a real bug (chart labels hardcoded to the pilot's sample counts) that every prior run had masked. It also hit a transient dataset-streaming error, auto-recovered, which validated moving the full study to a locally cached copy of the dataset (now implemented) rather than streaming.

9 Honest Limitations

- **Scale:** 163 of 3,004 test images; single model; single GPU.
- `struck_by_risk` **scarcity is a labeling artifact, not a dataset property.** The mutually-exclusive priority scheme discarded real struck-by images that also had a higher-priority violation; a direct scan shows the true multi-label pool is ∼70 across the full dataset, not 13. Every `struck_by_risk` number in this report rests on n=13 and is directional only.
- **The proxy, not the model, makes the judgment** (Section 7), so some metrics partly measure geometric-rule sensitivity.
- **Ranking by area** is an acknowledged stand-in for a true importance signal.

- **Single-worker detection**, hardcoded 2D proximity thresholds, bounded (not full) completeness, Level-1 (plus partial Level-2) robustness only, an uncalibrated confidence proxy, and an n=4 adversarial stretch test that is a plausibility probe, not a result.
- **Decoding was fragmented** across metrics; the full study should unify it.

10 Scale-Up Direction

The recommended scale-up is *not* simply "run 3,004 instead of 163." Scaling a biased pipeline only yields a larger biased sample. The plan (detailed separately) is phased:

1. **Foundational fixes first** — rule-aware region ranking; multi-label class labeling (which recovers `struck_by_risk`); bake the worker-loss correction into every metric; split the rule_2 prompt concepts; unify decoding. Cheap, high-leverage, unblocks everything.
2. **Metric-validity fixes** — size-invariant sparsity and stability; text-token ablation for descriptive accuracy; severity sweeps for robustness; multi-hazard completeness.
3. **Reproducibility and statistics** — containerized environment, multi-seed runs, minimum-n floors and nonparametric significance tests, programmatically-generated report numbers.
4. **Attribution** — keep cross-attention, add integrated gradients as a corroborating second stream.
5. **Scope** — add a native-VQA model (Qwen2.5-VL, in same-proxy *and* native-VQA comparisons, so the metrics measure model faithfulness), scale ConstructionSite to the full split, and extend to the SODA (out-of-distribution) and CMA (temporal explanation-stability) datasets.

Six decisions requested of Professor Abdallah

1. Is decoder→encoder cross-attention an acceptable stand-in for LRP/DeepLIFT, or is a transformer-native method required?
2. Minimum-n and statistical-testing standards, especially for the scarce `struck_by_risk` class.
3. Approval of the rule-aware region-ranking fix.
4. Full-dataset scaling vs. fixing class imbalance first.
5. Scope for a properly powered adversarial-robustness study.
6. Treating rule_2's two phrasings as distinct physical concepts.

11 Conclusion

The six-metric framework transfers from tabular models to a construction-safety VLM: all six metrics ran end-to-end and produced interpretable, traceable signal, after two adaptations that a feasibility pilot exists to surface. Just as importantly, the pilot identified exactly what must be fixed before the numbers can be trusted at scale — and, in doing so, produced a clear, evidence-backed engineering plan for a full study whose central aim is to make the metrics measure the model's reasoning rather than the scaffolding around it.